

Dr. ASHWIN SUDHAKARAN

Physics – M.Sc., Ph.D., Post-Doc.,
DOB: 11/09/1996 ; **Nationality:** Indian

ABOUT ME

I am a dedicated physicist with a strong focus on material science, passionate about both research and education. With a solid academic background and notable achievements (Gold medal), I create an inclusive learning environment with skills for academic success and beyond. I contribute to cutting-edge research and aim to inspire a lifelong love of learning and innovation, both as a researcher or an educator.

CONTACT

Phone

+91 90039 95469

+91 95004 39546

Email

dr.ashwinsudhakar@gmail.com

dr.ashwinoffice@gmail.com

ashwinsudhakar11@gmail.com

Address

4/425, Aravind Nagar 7th St.,
Thirunagar 3rd stop,
Thiruparankundram (T.K),
Madurai - 625006, TN, India

EXPERTISE

- ✓ Material Science
- ✓ Multifunctional Materials
- ✓ Nanocomposites
- ✓ Electrochemistry, etc.

RESEARCH RECORDS

Journal Publication -18
Conference Publication - 03
Presentations - 27
Projects undertaken - 03
Webinars - 68 (A), 13 (P)
Book Publication - 09
Workshop - 04 (A), 02 (O)
FDP - 08 (A), 04 (O)
Int'l Conference – 01 (O)
Research Conclave – 01 (O)
Certification Course - 04 (A)
PDP - 03

**A = Attended, O = Organized*
P = Presented

LANGUAGE

English, Tamil, Hindi

EXPERIENCE & SKILLS

Research

- ✓ 7+ years doctoral and postgraduate research on nanoscale characterization of nanocomposites using advanced microscopy and spectroscopy
- ✓ Expertise in specialized equipment - Scanning probe (SPM-AFM), Magnetic Force Microscopy (MFM), FTIR, XRD, SEM, UV (Vis), PL, DSC Thermal Vacuum Evaporation unit (Thin Film coating), CH (1760e), GC (1100), TGA, and STA.
- ✓ Scientific Software Used: Full-Prof, MAUD, GSAS-II, JANA 2006, HighScore Plus, Match-3, Profex, Pcpdfwin, VESTA, ChemDraw Ultra, Ortep, WInGX, OMNIC, IRPal, AutoCAD, Fusion360, FreeCAD, ImageJ, Origin Lab, MATLAB, QtiPlot, Wien2k, COMSOL, VINUS, OOMMF, 3D CAD modeling and simulation: AutoCAD, FreeCAD, Fusion 360, and Turnitin.

Responsibilities

- ✓ Reviewer - Journal of Sol-Gel Science and Technology, Optical and Quantum Electronics, Journal of Electroceramics - (Springer Nature). Chemistry Select (Wiley). (2024 - present)
- ✓ Associate Co-Editor & Reviewer - Scientific Reports (Nature Portfolio). (2024 - present)
- ✓ @SRMMCET (2024) : Former. Head for R&D Cell -S&H, Social media Influence & Joint coordinator of Helix (Science Club).
- ✓ @TCE (2025 - present) : Department level Coordinator - Centre of Excellence in Materials Research and Energy Solutions (CEMRES) & Extra-curricular activities, 1st Yr BE Students coordinator, BE Lab, NAAC/NIRF/NBA, Bill passing, Stock & General

Teaching & Mentoring

- ✓ Actively mentored young scholars through academic projects (UG & PG: 2020- 2023) and BE (2024-2025).

Internship : Specialized research interns & PG project in nanomaterials @IGCAR, Kalpakkam.

Membership: IFERP, NDLI, Lifetime Member of IAENG, NITTSD and AICTSD

EDUCATION

2024 – 2025, *CNMC, Jain University* **Post Doctoral Fellow**	Independent researcher in the field of Material Science and Chemistry, on the topic of synthesis of bi-metallic and electrospun ternary transition metal-oxide nanofiber catalysts for electrocatalytic CO_2 reduction and hydrogen evolution reactions (HER).
2019 – 2024, *Karpagam University* **Ph.D., Physics (Materials Science)**	Researched in Material Science, focusing on synthesizing Ferrite nanocomposites and characterizing multifunctional physical properties for magnetism and photonics applications.
2017 – 2019, *Karpagam University* **M.Sc., Physics**	Focused on synthesizing phase-pure nanostructures and characterizing them using advanced microscopy techniques (AFM & MFM), unveiling insights into magnetic domain behavior and superparamagnetism
2014 – 2017, *Karpagam University* **B.Sc., Physics**	Focused on physics coursework, achieving high academic performance, and demonstrating exceptional understanding of core concepts.

References.,

1. **Dr. A. Saranya,** Assistant Professor, Vel Tech University, Chennai, TN, India. saran.amirth@gmail.com
2. **Dr. Arvind H Jadhav,** Associate Professor, CNMS, Jain University Global Campus, Bangalore, India, j.arvind@jainuniversity.ac.in
3. **Dr. S. Arumugam**, Vice Chancellor, Tamil Nadu Open University, Chennai-600015, TN, India. sarumugam1963@gmail.com
4. **Dr. G. Mangamma,** Associate Professor, HBNI & Scientific Officer H, Head, NCSS-SND, IGCAR, Kalpakkam, India. gm@igcar.gov.in
5. **Dr. P. Pandaram,** Scientific Officer H, TLD Lab, Kudakulam Nuclear Power project, Kudakulam, TN, India, p.pandaram@gmail.com